Unsupervised CD in Satellite Image Time Series by Contrastive Learning and Feature Tracking

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***Abstract*— Unsupervised change detection using contrastive learning has significantly improved the performance of literature techniques. However, at present it only focuses on the bi-temporal change detection scenario. Previous state-of-the-art models for image time-series change detection have traditionally depended on features obtained either through clustering learning or by training models from scratch using pseudo labels tailored to each scene. However, these approaches fail to either exploit the spatial–temporal information of image time-series or generalize to unseen scenarios. In this work, we propose a two-stage approach to unsupervised change detection in satellite image time-series using contrastive learning with feature tracking. By deriving pseudo labels from pretrained models and using feature tracking to propagate them within the image time-series, we improve the consistency of our pseudo labels and address the challenges of seasonal changes in long-term remote sensing (RS) image time-series. We adopt the self-training algorithm with ConvLSTM on the obtained pseudo labels, where we first use supervised contrastive loss and contrastive random walks to further improve the feature correspondence in space-time. Then, a fully connected layer is fine-tuned on the pretrained multitemporal features for generating the final change maps. Through comprehensive experiments on two datasets, we demonstrate consistent improvements in accuracy on fitting and inference scenarios.**

***Index Terms*— Change detection, contrastive learning, feature tracking, multitemporal, remote sensing (RS).**

I. Introduction

DETECTION of changes in multitemporal remote sensing (RS) images has been extensively studied in the past decades [1]. Early approaches to change detection in bi-temporal RS images include image algebra, image transformation and image classification methods [2]. These methods have limitations, such as relying on empirical feature extraction algorithms or being sensitive to classification results, which limit their application in change detection. Image algebra methods directly compare image values, such as in the case of change vector analysis (CVA)-based methods [2], [3], [4], [5] that provide spectral change information in terms of magnitude and direction of the spectral change vectors. Image transformation methods map images into the same feature space for comparison. The most common transformation methods include principal component analysis (PCA) [6], slow feature analysis (SFA) [7], and canonical correlation analysis (CCA) [8]. Supervised image classification methods project image values into different classes at each date and compare directly class labels. This approach, known as postclassification comparison [1], is widely used in large-scale land-cover change detection. In general, image algebra and transformation methods heavily rely on empirical feature extraction algorithms, while postclassification methods are sensitive to the classification results of each image and to error propagation. These limitations hinder the application of conventional change detection methods.

Deep learning methods have been shown to significantly improve the performance of conventional change detection methods by using deep neural networks [9] and stochastic gradient descent [10]. One common approach is direct classification, where models are trained using predefined labels and then used to classify changed and unchanged pixels. For example, Daudt et al. [11] presented three Unet-based convolutional neural network (CNN) architectures for detecting binary changes between pairs of registered RGB images. In the absence of ground truth, pseudo labels from conventional change detection methods can be used to train models in a self-training paradigm. Zhou and Li [12] proposed a self-training algorithm based on pseudo labels for change detection, where the pseudo labels are generated by the traditional CVA approach and used to train an end-to-end new network. The image transformation approach has also been improved using deep learning, where deep neural networks are utilized to extract discriminative features.

Many new techniques have been developed for extracting discriminative features from bi-temporal RS images, based on both generative [13] and discriminative [14] models. For the generative model, Luppino et al. [15] combined domain-specific affinity matrices and autoencoders (AEs) to align related pixels from multimodal images. Chen and Bruzzone [16], [17] explored the use of discriminative models in change detection, proposing the use of contrastive learning at pixel-level and patch-level in multitemporal and multisensor scenarios. The application of deep learning in postclassification comparison change detection can follow two main directions. One is to use a deep learning-based segmentation approach to classify the object of interest on bi-temporal images and then compare them. For example, Nemoto et al. [18] first segment buildings in an urban area

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and then compare the building maps at two different times to detect changes. Another approach is to perform binary change detection and segmentation of both images simultaneously. Ding et al. [19] proposed combining postclassification and direct classification methods using a bi-temporal semantic reasoning network, where the network produces both a change map and two classification maps. These approaches demonstrate the ability of deep learning in deriving pixel-wise changes from image pairs. In addition to pixel-wise change detection, large-scale scene change detection is also explored for helping urban planning [20], [21], [22].

The challenge of detecting changes in RS images time-series is compounded by the presence of seasonal noise, which can be difficult to distinguish from true changes. One approach to address this challenge is to use graph-based methods [21], [23], which present detected spatio-temporal phenomena as evolution graphs composed of spatio-temporal entities belonging to the same geographical location in multiple timestamps. Deep learning methods have also been applied to RS image time-series change detection, using techniques such as recurrent neural networks (RNNs) [24] to extract discriminative features from image sequences. However, the supervised RS image change detection methods are constrained by the diversity and quantity of available training samples, which is more challenging for RS image time-series change detection tasks. Current supervised RS change detection methods concentrate principally on a restricted number of land-use and land-cover (LULC) classes, and are trained using bi-temporal change maps. To overcome this limitation and incorporate a broader scope of change types, as well as to extend the applicability to RS image time-series change detection, researchers have explored the utility of self-training paradigms, including self-supervised learning and pseudo-label learning techniques. These methods entail training networks on pretext tasks—such as image reconstruction using 3-D CNNs [25], [26] and predicting the correct order of shuffled image sequences [27]. For example, Kalincheva et al. [28] proposed a framework combining a graph model and pseudo labels, which associates changes in consecutive images with different spatial objects using a gated recurrent unit (GRU) AE-based model. Meshkini et al. [25] further proposed the use of a pretrained 3-D CNN to extract spatial–temporal information from long satellite image time-series, where they can detect the times and locations of changes in image sequences. Nonetheless, the algorithm based on feature learning [25] continues to depend on a thresholding technique to generate final change maps, which inevitably incorporate noise. Moreover, the pseudo labels utilized in self-training [28] are frequently contaminated with high levels of noise and neglect the temporal dynamics of changes.

In this context, the present study endeavors to integrate the robust pretrained features derived by self-supervised learning with the thresholding-based pseudo labels to mitigate the impact of label noise, which enables the neural network to distinguish between changed and unchanged features. By adopting this strategy, the research aims to avoid the noise inherently associated with thresholding and to bypass the labor-intensive process of manually annotating a diverse semantic scope within lengthy image sequences. In addition, to further reduce the effects of label noise and to build the temporal correlations between changes during network training, we further employ the integration of supervised contrastive learning and the contrastive random walk methodology. Specifically, in this work, we propose the use of contrastive learning [29] and feature tracking [30] to address these challenges and improve the performance of change detection in RS image time-series. We leverage contrastive learning methods both to get good pretrained features for pseudo label generation and to reduce the overfitting that results in incorrect pseudo labels when considering supervised contrastive learning [29] and contrastive random walk [31]. Additionally, by incorporating a feature tracking-based pseudo label generation task and a convolutional long–short-term memory network (ConvLSTM) [32], we are able to extract time-series change maps from image time-series and further train a new model from scratch. In detail, the pseudo-label generation is based on the pretrained model using contrastive learning. The change detection model is trained from pseudo labels by the joint use of Unet [33] and ConvLSTM networks. We first extract pseudo labels from change pair time-series and then use them with images to train the proposed network, which outputs change maps relative to the first image in the sequence. During the training, supervised contrastive loss, contrastive random walk loss, and logistic regression are used to optimize the parameters of the feature encoder and the last classifier, respectively. The supervised contrastive loss is used to mitigate the noise in pseudo labels, while the contrastive random walk loss improves the quality of the consecutive change results. Finally, we demonstrate the effectiveness of our approach on two datasets.

The approach proposed in this article contains the following main novel contributions.

1) To generate time-related pseudo labels for network training, we propose to use feature tracking to extract reliable change pixels in image sequences that are insensitive to seasonal noise.
2) To ensure the robustness and consistency of change maps, we propose to use supervised contrastive loss and contrastive random walk loss on change feature learning. These losses encourage the pixels in the same class to have a closer feature representation among image time-series.
3) To extend the approach to arbitrary long time-series, we jointly use Unet and ConvLSTM as the model architectures. To verify the performance of the proposed approach, we provide a comparison with state-of-the-art methods and an ablation study. Our experiments show that our method obtains competitive results on the datasets.

The remainder of this article is organized as follows. Section II introduces the related works. Section III presents our proposed approach, including the network architecture, the supervised contrastive loss, the contrastive random walk loss, and the feature tracking-based pseudo label updating. Section IV illustrates the experimental results obtained on two datasets and compares our approach to state-of-the-art

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methods. This section also reports the results of an ablation study. Finally, Section V draws conclusion and discusses future work.

II. Related Works

A. Self-Supervised Learning in Change Detection

Self-supervised learning is a method of representation learning that does not require human intervention for data annotation, as opposed to supervised learning. It has been successful in RS image change detection, thanks to its ability to obtain good representative features. Self-supervised learning has two main streams: generative and discriminative methods [14].

Generative models often rely on AEs, generative adversarial networks [34], and diffusion models [35]. Denoising AEs [36], a classical generative self-supervised learning model, are a type of AE that reconstructs one temporal image from another. Bergamasco et al. [37] proposed the use of a multilayer convolutional denoising AE for unsupervised change detection in multitemporal Sentinel-1 images. In addition to AEs, generative adversarial networks have also been used for change detection tasks. For example, Gong et al. [38] treated change detection as a generative learning procedure that connects bi-temporal images and generates change maps. As for diffusion models, Gedara Chaminda Bandara et al. [39] used a pretrained denoising diffusion probabilistic model to extract feature representations from unlabeled RS images, and then trained a lightweight change detection classifier to detect changes from the learned features.

In contrast, discriminative models are mostly based on contrastive learning, which learns a representation that helps distinguish one object from another. The objective of contrastive loss is to learn a representation where semantically similar features are brought closer together and dissimilar features are pushed apart. Unlike fine-tuning on downstream tasks, unsupervised change detection using well pretrained features relies on thresholding approaches and only considers bi-temporal images. Discriminative models used in self-supervised change detection include predefined tasks and contrastive methods. Leenstra et al. [40] used predefined tasks for feature representation learning and trained a discriminative model to extract features from bi-temporal images for change detection. An early attempt to use contrastive methods in change detection is [17], where the authors proposed contrastive learning for change detection in multiview RS images (including multitemporal and multisensor images). In [16], they further proposed a pixel-wise contrastive approach to distill the features to alleviate seasonal effects in bi-temporal change detection. One challenge that these works face in using contrastive approaches to unsupervised pretraining is that they push apart samples that should belong to the same class, making it harder for the classifier to later categorize them correctly or create accurate decision boundaries.

B. Change Detection in RSI Time-Series Using ConvLSTM

Change detection is often associated with sequential data, making it necessary to evaluate temporal dynamics. The computer vision community has addressed the modeling of temporal relationships among features using RNNs, which have proven effective for a wide range of applications such as object tracking and action recognition. Long short-term memory (LSTM) networks are particularly effective for such problems, as they mitigate the vanishing gradient problem when dealing with long-term dependencies. The combination of RNNs and deep learning architectures has been used for time-series tasks, aiming to produce more useful feature representations by extracting both spatial and temporal information during the learning process. Recent RS image time-series change detection tasks have extensively integrated LSTM techniques. In [41], an LSTM is integrated into a CNN to consider both spatial and temporal features in an end-to-end framework. Sefrin et al. [42] proposed combining FCN and LSTM to study land-cover changes using Sentinel-2 images. For high-resolution image change detection, Sun et al. [43] proposed using atrous Unet-ConvLSTM to better model multiscale spatial information. For unsupervised approaches, Saha et al. [27] treated change detection as an anomaly detection problem, using an LSTM network to learn a representation of the image time-series. In this method, they used a pretext task of reordering the image sequence. However, the predefined task cannot resist the influence of seasonal noise, which leads to many pseudo-changes in the results. Some researchers have shown that pseudo labels can help solve this problem. Kalinicheva et al. [28] proposed a framework that combines a graph model and pseudo labels, using a GRU AE-based model to associate the changes of consecutive images with different spatial objects. Yang et al. [44] proposed an unsupervised time-distance-guided convolutional RNN (UTRnet) for change detection in irregularly collected images, using a weighted prechange detection to obtain reliable training samples.

III. Methodology

In this section, we present the proposed two-stage RS image time-series change detection framework. It includes a feature tracking-based pseudo label generation task and a self-training change-detection module that follows the training setting of supervised contrastive learning [29]. We first get the pixel-wise feature representation of each image in the image sequence using the pretrained model [16] and then get the pseudo-change maps using the thresholding approach. Then, the feature tracking approach is used to update the threshold-based pseudo-change maps. Afterward, the pseudo-change labels are used to learn the representation of change maps using the supervised contrastive loss and the contrastive random walk loss. Finally, a fully connected layer is fine-tuned on the learned change map representation using logistic regression. In Sections III-A–III-C, we will describe the network architecture of the proposed framework, the supervised contrastive loss, the contrastive random walk loss, and the feature tracking-based pseudo label updating.

A. Network Architecture

The proposed approach uses a Unet-ConvLSTM network architecture, which consists of two components: ResUnet and

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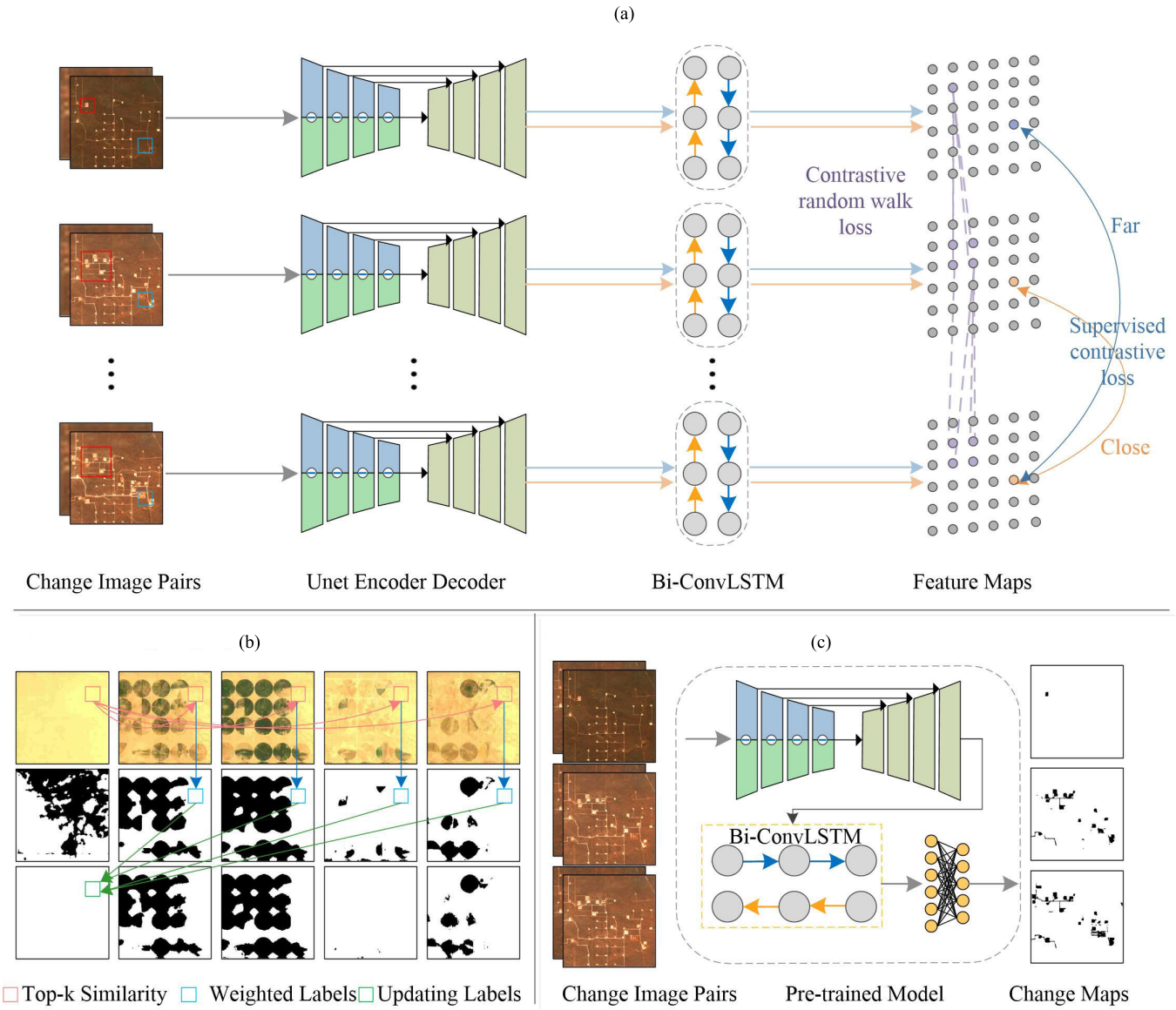


Fig. 1. Overview of the proposed approach for RS image time-series change detection, where the proposed network is based on the Unet and Bi-ConvLSTM. (a) Pretraining step uses supervised contrastive learning on spatial feature representation and uses contrastive random walk loss on temporal features for temporal feature modeling. (b) Label propagation step uses k-NN for noise reduction among change map time-series. (c) Fine-tuning step uses an MLP and logistic regression to predict the final change maps.

Bi-ConvLSTM (Fig. 1). For the ResUnet, we adopt a similar architecture as the FC-Siam-diff [11]. Instead of concatenating features from two encoders, it instead uses the absolute value of their feature difference. It consists of two encoders, one bridge, one decoder, and skip connections between the down-sampling and upsampling paths. The decoder part has three blocks, each of which consists of a convolution layer (Conv), batch normalization (BN), ReLU, and upsampling. A 1×1 Conv is used after the last block to reconstruct the learned representations. We changed the padding type of all blocks to "same" padding. The parameters and channel size of each unit are presented in Table I. Each convolution unit ($[\cdot]$) includes a convolutional layer, a BN layer, and a ReLU activation layer. Each residual block (ResBlk) in the encoding path has two residual units, each of which consists of two convolution units and an identity mapping.

The output features of time-series change pairs are given in the input to the Bi-ConvLSTM layer. Different from the standard LSTM, ConvLSTM uses convolution operations in the input-to-state and state-to-state transitions to improve the modeling of the spatial correlation among sequence images. It consists of an input gate i_t, an output gate o_t, a forget gate f_t, and a memory cell $\mathcal{C}_t$. The input–output and forget gates act as controlling gates to access, update, and clear memory cell. ConvLSTM can be formulated as follows (for convenience we remove the subscript and subscript from the parameters):

$$\begin{aligned} i_t &= \sigma(W_{xi} X_t + W_{hi} H_{t-1} + W_{ci} C_{t-1} + b_i) \\ f_t &= \sigma\left(W_{xf} X_t + W_{hf} H_{t-1} + W_{cf} C_{t-1} + b_f\right) \\ C_t &= f_t \circ C_{t-1} + i_t \tanh(W_{xc} X_t + W_{hc} H_{t-1} + b_c) \\ o_t &= \sigma(W_{xo} X_t + W_{ho} H_{t-1} + W_{co} \circ C_t + b_c) \\ H_t &= o_t \circ \tanh(C_t) \end{aligned} \tag{1}$$

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TABLE I
STRUCTURE OF THE PROPOSED NETWORK

	Encoder 1 & 2	Decoder	
Conv1 Maxpool	[3×3, 32], stride 2 3×3, stride 2	Cat. Diff. ResBlk2 [3×3, 64] upsampling 2	DecBlk1 stride 1
ResBlk1 stride 1	$\begin{bmatrix} 3\times3,\ 32 \\ 3\times3,\ 32 \end{bmatrix}\times 2$	Cat. Diff. ResBlk1 [3×3, 32] upsampling 2	DecBlk2 stride 1
ResBlk2 stride 2	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times 2$	Cat. Diff. Conv1 [3×3, 32] upsampling 2	DecBlk3 stride 1
ResBlk3 stride 2	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times 2$	[3×3, 16] Bi-ConvLSTM	LSTM Blk
Bridge stride 1	[3×3, 128] upsampling 2	[1×1, 32/1]	Logistic Regression

where o denotes the Hadamard functions. X_t is the input tensor and H_t is the hidden state tensor. W_{x*} and W_{h*} are 2-D convolution kernels corresponding to the input and hidden state, respectively, and b_i, b_f, b_o, and b_c are the bias terms. In this study, we employ Bi-ConvLSTM [45], [46] to encode the features of time-series change pairs. It was proposed to use both past and future information to model sequential data. Bi-ConvLSTM uses two ConvLSTMs to process the input data in both forward and backward directions and then makes a decision for the current input by taking into account the data dependencies in both directions. It has been shown that analyzing both forward and backward temporal perspectives improves predictive performance. Each forward and backward ConvLSTM can be considered as a standard one, with two sets of parameters for backward and forward states. The output of Bi-ConvLSTM is calculated as follows:

$$\mathbf{Y}_t = \tanh\left(\mathbf{W}_y^{\overrightarrow{H}} \overrightarrow{H}_t + \mathbf{W}_y^{\overleftarrow{H}} \overleftarrow{H}_t + b\right) \tag{2}$$

where $\overrightarrow{\mathcal{H}}_t$ and $\overleftarrow{\mathcal{H}}_t$ denote the hidden state tensors for forward and backward states, respectively, b is the bias term, and $\mathbf{Y}_t$ indicates the final output considering bidirectional spatio-temporal information. The hyperbolic tangent (tanh) is used to combine the output of both forward and backward states in a nonlinear manner. After the last layer of Bi-ConvLSTM, an MLP block is used to reconstruct output features at the feature learning stage and predict the binary change maps at the fine-tuning stage.

B. Loss Function

During training, each image $I_i (i > 0)$ is used to construct a change pair anchored at the initial image ($I_{i=0}$). In this way, the proposed network captures the temporal changes related to the first image rather than the cumulated changes of the image sequence. The training process uses a teacher–student paradigm and the exponential moving average (EMA) algorithm [47]. The inputs of the student network are the original time-series image pairs, while the teacher network uses the same time-series image pairs with color jitter.

According to supervised contrastive learning, the training process consists of the feature learning and fine-tuning stages. In the feature learning stage, we use supervised contrastive loss [29] with the contrastive random walk loss [31]. The loss can be written as

$$\mathcal{L}_{\text{feat}} = \mathcal{L}_{\text{sc}} + \lambda \mathcal{L}_{\text{crw}} \tag{3}$$

where $\mathcal{L}_{\text{sc}}$ is the supervised contrastive loss and $\mathcal{L}_{\text{crw}}$ is the contrastive random walk loss. In the fine-tuning stage, we use logistic regression to directly predict the change probability. The hyper-parameter λ is used to tune the loss. A value of $\lambda = 0.1$ generally performed well in our experiments.

Image time-series change detection is often treated as a simple extension of bi-temporal change detection in time. The supervised contrastive loss is used to differentiate representations between changed and unchanged pixels spatially in time-series change pairs. However, the incorporation of temporal information to mitigate seasonal noise poses a significant challenge because the change depicted in frame t might not have any relation to what we find at the same location in frame $t + k$. To overcome this limitation, the contrastive random walk leverages pathfinding on a space-time graph and associates features across space and time. It establishes feature correspondence shared by neighboring frames. In the following, we provide details on the supervised contrastive loss and the contrastive random walk loss.

1) Contrastive Loss: The supervised contrastive loss is calculated by sampling the pixel features in the constructed time-series pairs. The pixel feature pairs at the same location in the output of the teacher and student networks are called positive pairs, while pixel features from different locations are called negative pairs. Given a positive feature pair (v_1^i, v_2^i) and a pixel feature v_2^j taken from another location, the contrastive loss can be formulated as $\mathcal{L}_{\text{contrast}}$

$$\mathcal{L}_{\text{contrast}} = -\mathop{\mathbb{E}}_{S}\left[\log \frac{e^{\text{sim}(v_1^i, v_2^i)/\tau}}{\sum_{j=1}^{N} e^{\text{sim}(v_1^i, v_2^j)/\tau}}\right] \tag{4}$$

where sim is a similarity function (i.e., cosine similarity), (v_1^i, v_2^i) is the normalized latent representation of pixel i, $(v_1^i, v_2^j | j \geq i)$ is the normalized latent representation of negative pair and $S = \{s_1^1, s_2^1, s_2^2, \cdots, s_2^{N-1}\}$ is a set that contains $N-1$ negative samples and one positive sample. One limitation of self-supervised contrastive learning is that, since the class labels of the inputs are ignored, samples from the same class may end up being treated as negative pairs, which can affect the training performance. To avoid this limitation and enable the contrastive loss to learn in a supervised fashion, Khosla et al. [29] extended the approach to account for input labels. Following the original supervised contrastive learning method, we randomly sample N pixel features in each change pair from the teacher–student network, generating two data views $\{(x_i, y_i)\}_{i=1}^{2N}$, where $i \in I = [2N]$ is the index of an arbitrary sample. Given $A = \{A_{i,j} | y_i = y_j, (x_i, y_i), (x_j, y_j)\}$, we perform supervised contrastive learning with sampled pixel features

$$\mathcal{L}_i^{\text{sup}} = \sum_{i \in I} \frac{-1}{|A(i)|} \cdot \sum_{a \in A(i)} \log \frac{e^{\text{sim}(x_i, y_a)/\tau}}{\sum_{b \in B(i)} e^{\text{sim}(x_i, y_b)/\tau}} \tag{5}$$

where $B(i)$ means the set of indices excluding i, i.e., $B(i) = I \backslash i$; $A(i) = \{A_{i,j} | j \in B(i)\}$ is the positive set distinct from

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sample i and $|\cdot|$ stands for cardinality. In this case, the labels are binary pseudo labels.

2) Contrastive Random Walk Loss: This work builds upon the contrastive random walk framework by Jabri et al. [31]. In the contrastive random walk, we are given an input image time-series with k frames. We select N nodes q_t from the frame t, which serve as vertices of a graph. Pairwise similarities of nodes are converted into nonnegative affinities by applying a softmax function (with temperature τ) over edges departing from each node. This process generates the stochastic affinity matrix between frames t and $t+1$ as a bipartite graph with each edge given as

$$C_t^{t+1}(i,j) = \text{softmax}(q_t q_{t+1}^T)_{i,j}. \tag{6}$$

The affinity matrix for the entire graph denotes the edge weights between all pairs of nodes in change image pairs. To model affinities over multiple change image pairs, we take the product of the sequential affinity matrices

$$\overline{C}_t^{t+k} = \prod_{i=0}^{k-1} C_{t+i}^{t+i+1}. \tag{7}$$

Ultimately, we train the model to maximize the likelihood of cycle consistency, i.e., the event that the walker returns to the node it started from

$$\mathcal{L}_{\text{crw}} = -\text{tr}(\log(\overline{C}_t^{t+k}\overline{C}_{t+k}^{t})). \tag{8}$$

C. Pseudo Label Updating

Bi-temporal RS image change detection usually gets the change map through thresholding methods, which inevitably introduce errors when there is no change between bi-temporal images. The pseudo-change maps can be made less noisy by propagating the threshold-based pseudo labels to each change pair using the label propagation algorithm [30]. This is because the false alarms in one frame can be mitigated by similar features with correct labels from other frames. This algorithm propagates labels in the feature space considering both spatial and temporal neighbors. In detail, the labels of target nodes are determined by computing the matrix of transitions between target nodes and source nodes, considering only the top$-k$ transitions, and multiplying it by the labels of the source nodes. Given the feature embedding of a frame I_t and a one-hot format label of the frame I_{t-1}, we compute its cosine similarity with the feature embedding of the frame I_{t-1}

$$M_{t-1,t} = f(\phi(I_{t-1}), \phi(I_t)). \tag{9}$$

Then, we compute the label y_i of the pixel i in I_t according to the label of pixel j in I_{t-1}

$$y_i = \sum_j M_{t-1,t}(j,i) y_j \tag{10}$$

where $M_{t-1,t}(j,i)$ is the affinity between pixel i of I_t and pixel j of I_{t-1}, and ϕ is the encoder. For each pixel i, we propagate from the top N pixels with the greatest affinity $M_{t-1,t}(j,i)$ for each pixel i. For example, let us assume we get a false alarm f_{ij} at the pixel $(i,j)_t$ of the frame t. Then we compute its cosine similarity with the feature of pixel $(i,j)_{\setminus t}$ from the remaining frames in the time series and select the $N = 10$ feature embeddings with the highest similarity. We then use these embeddings to compute a weighted sum of the label predictions at the pixel $(i,j)_t$ of frame t. This process is repeated for all change pairs in the image sequence, updating the labels and embedding contexts using the top-N change pairs. Finally, the false alarms in the frame $(i,j)_t$ are corrected by the pseudo labels from other frames.

IV. Experiments

In this section, we first describe the datasets used in our experiments and then introduce the related experiment setting on the network training and the pseudo label updating. Finally, we present the results of the proposed approach and the comparison methods. We also present an ablation study of each component of the proposed approach and the hyperparameter in the loss function.

A. Description of Datasets

We conducted experiments on two multispectral datasets, one from the Sentinel-2 satellite constellation and the other from the Landsat-8 satellite.

1) Sentinel-2 Dataset: The multitemporal urban development (MUDS) dataset [48] was designed to monitor urbanization by tracking changes in building construction from 2017 to 2020. It is an open-source dataset that includes native Planet 4-m resolution imagery and Sentinel-2 multispectral images with irregular observation intervals across six continents. However, the original Sentinel-2 images often contain clouds and missing values. To improve the utility of this dataset, we selected only 74 locations with a minimum of 12 clean images and resized each image to 512 × 512 pixels. Of these 74 locations, we labeled the significant change pair of 53 scenes for future evaluation, with all change pairs referenced to the first image. We used only four bands in this work, all of them with a spatial resolution of ten meters. Due to the unsupervised nature of this dataset, we only considered three types of changes: built-up, bare land, and water.

2) Landsat-8 Dataset: The UTRnet dataset [44] was specifically designed for validation of the UTRnet model. The dataset consists of the satellite image time-series collected by Landsat-8 from 2013 to 2021, with a spatial resolution of 30 m. Four spectral bands covering the visible to the shortwave infrared region are used, including blue, green, red, and near-infrared bands. The dataset includes nine typical scenes located in different cities in China, each with a different land-cover type. For each scene, ten cloud-free Landsat-8 images were selected to cover different seasons. The image size for each scene is 400 × 400 pixels. The ground truth includes three classes: changed pixels, unchanged pixels, and unlabeled pixels. The changed and unchanged pixels are labeled using Google Earth images. In this study, unlabeled pixels are treated as unchanged pixels to validate the influence of seasonal noise. Due to the temporal limitations of high-resolution image labeling, the labels only include the longest interval pairs. The change maps include city expansion, water change, and soil change.

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B. Experiment Settings

1) Evaluation Metrics: In order to evaluate the effectiveness of different methods in binary change detection, this article employs five evaluation metrics: precision (Pre), recall (Rec), overall accuracy (OA), $F1$ score ($F1$), and Cohen's kappa score (Kap).

2) Implementation Details: In the process of generating pseudo labels, we first derived pseudo labels of each change pair using a thresholding approach on pretrained features [16]. Then, propagate the threshold-based labels to each change pair using the feature tracking approach. In the setting of feature tracking parameters, the spatial neighbors P are set to 10, the temporal neighbors N_T are set to 3 most correlated change pairs, and the top_k pixel is set to 10. In the self-training algorithm, the proposed approach uses a two-layer Bi-ConvLSTM. We choose the Adam optimizer with an initial learning rate of $3e^{-4}$ at the feature learning stage, which is decreased using step scheduling without restarts. The batch size is set to 2 and the model is trained for 200 epochs. For the fine-tuning, we use an SGD optimizer with a learning rate of 0.01, a mini-batch size of 10, and a number of epochs equal to 10. To evaluate the proposed approach, it is compared with the state-of-the-art method UTRnet in fitting and unseen scenarios. UTRnet is an improved LSTM-based self-training approach that uses CVA to generate pseudo labels. Unlike the proposed approach, UTRnet is not designed to generalize to unseen scenarios and requires fitting a separate model for each scene. In addition to the image time-series change detection approaches, we also considered three state-of-the-art unsupervised multitemporal image change detection approaches [15], [16], [49]. The PixSSL change detection approach [16] exploits the contrastive loss and uncertainty approach to align multitemporal images. The CAA [15] is an AE-based generative model and the GMCD [49] is based on graph convolutional network (GCN) and metric learning.

In the evaluation, we chose the fitting evaluation on the Landsat-8 dataset due to the lack of training data, whereas considered the inference evaluation on the Sentinel-2 dataset. For the fitting scenarios of the Landsat-8 dataset, we chose Scenes 3, 5, and 7 as the evaluation set. For the unseen scenarios of the Sentinel-2 dataset, we chose scene $T1286_2921_13$, scene $T1736_3318_13$, and scene $T6730_3430_13$ as the evaluation set. In addition to the comparison with UTRnet, we also conduct extensive ablation experiments on the labeled 53 scenes of the Sentinel-2 dataset to evaluate the impact of different components of the proposed approach and using different pseudo labels. In particular, the proposed approach is compared with its versions that do not use the contrastive random walk loss or only use logistic regression directly. It should be noted that only the change pair with the most significant change is labeled as ground truth for evaluating the performance of different approaches.

C. Experimental Results on Landsat-8 Image Time-Series

In this study, the effectiveness of the proposed approach is evaluated using the Landsat-8 dataset. The performance of the proposed approach is compared with the state-of-the-art approach UTRnet, which has been validated by fitting on each scene in the dataset. In order to evaluate the generalization capability of the proposed approach, results are provided for fitting on all scenes, while UTRnet results are provided for both fitting on each scene and fitting on all scenes. Quantitative evaluation is performed on the most significant change map of the change map time-series, due to the challenges of differentiating changes in continuous change scenarios. The results of the proposed approach and comparison methods are presented in Table II. Among all bi-temporal image change detection approaches, the PixSSL obtained a Kappa coefficient of 0.557, which outperforms the results obtained by GMCD and CAA. This is because the PixSSL used the uncertainty approach to further suppress the seasonal noise among multitemporal images with respect to the only use of the self-supervised learning method. In the one-scene fitting setting, UTRnet achieves an OA of 89.66% and a Cohen's kappa score of 0.557, underperforming the results obtained from the pseudo labels. However, for the all-scene fitting setting, UTRnet fails to differentiate changed and unchanged pixels, achieving an OA of 48.50% and a Cohen's kappa score of 0.09. In contrast, the proposed approach achieves significantly better results than UTRnet in both settings, with an OA of 93.15% and a Cohen's kappa score of 0.692. Comparing the results of bi-temporal image change detection approaches, we can see that self-training approaches further improve the results of the bi-temporal change maps by using pseudo labels.

We also provide a visual comparison of the results obtained by the proposed approach and the UTRnet method. We present the results of UTRnet obtained by both fitting on each scene and on all scenes, as well as the results of the proposed approach. We also present the most significant change maps in each first column of the change maps in each scene in Fig. 2, where true positives, true negatives, false positives, and false negatives are colored in green, white, blue, and red, respectively. From the visual comparison of the most significant change map, we can see that the change map obtained by UTRnet using all-scene fitting is noisy and contains a high number of false alarms. In contrast, the change maps obtained by the other two settings are more accurate and have less noise. In addition, the proposed approach is able to successfully detect most of the changed pixels and suppress the effects of seasonal changes. When comparing the change map time-series, we can see that the change maps obtained by UTRnet (one-scene fitting) have more false alarms that are affected by historical changes. In contrast, the change maps obtained by the proposed approach are robust to seasonal changes and only focus on real changes that happened at each time. While the one-scene fitting UTRnet still achieves good results on all test scenes, the all-scene fitting UTRnet can perform well rarely in a few scenes with less seasonal noise, but its results are heavily influenced by the imbalanced training samples.

D. Experimental Results on the Sentinel-2 Image Time-Series

The Sentinel-2 dataset is characterized by a diversity of land-cover scenes and a larger number of training samples. In contrast to the results obtained on the Landsat-8 dataset,

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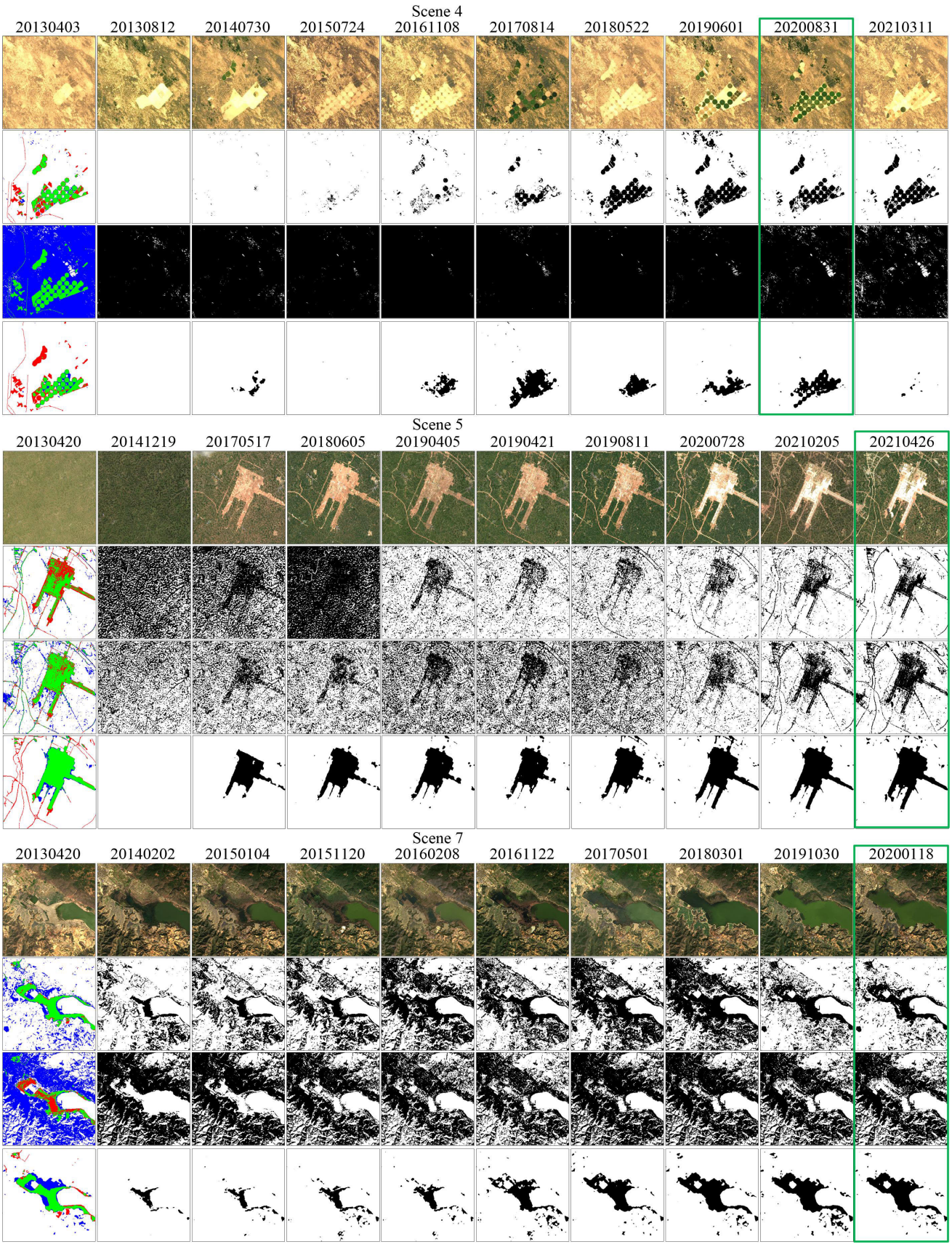


Fig. 2. Examples of change detection results on three scenes for the Landsat-8 dataset. Row 1: image time-series. Row 2: change maps of one-scene fitting obtained by UTRnet. Row 3: change maps of all-scene fitting obtained by UTRnet. Row 4: change maps of all-scene fitting obtained by the proposed approach. Column 1 of Rows 2–4 in each scene is the most significant change map versus the ground truth (green: TP, white: TN, blue: FP, and red: FN). The green box indicates the most significant changed image pair.

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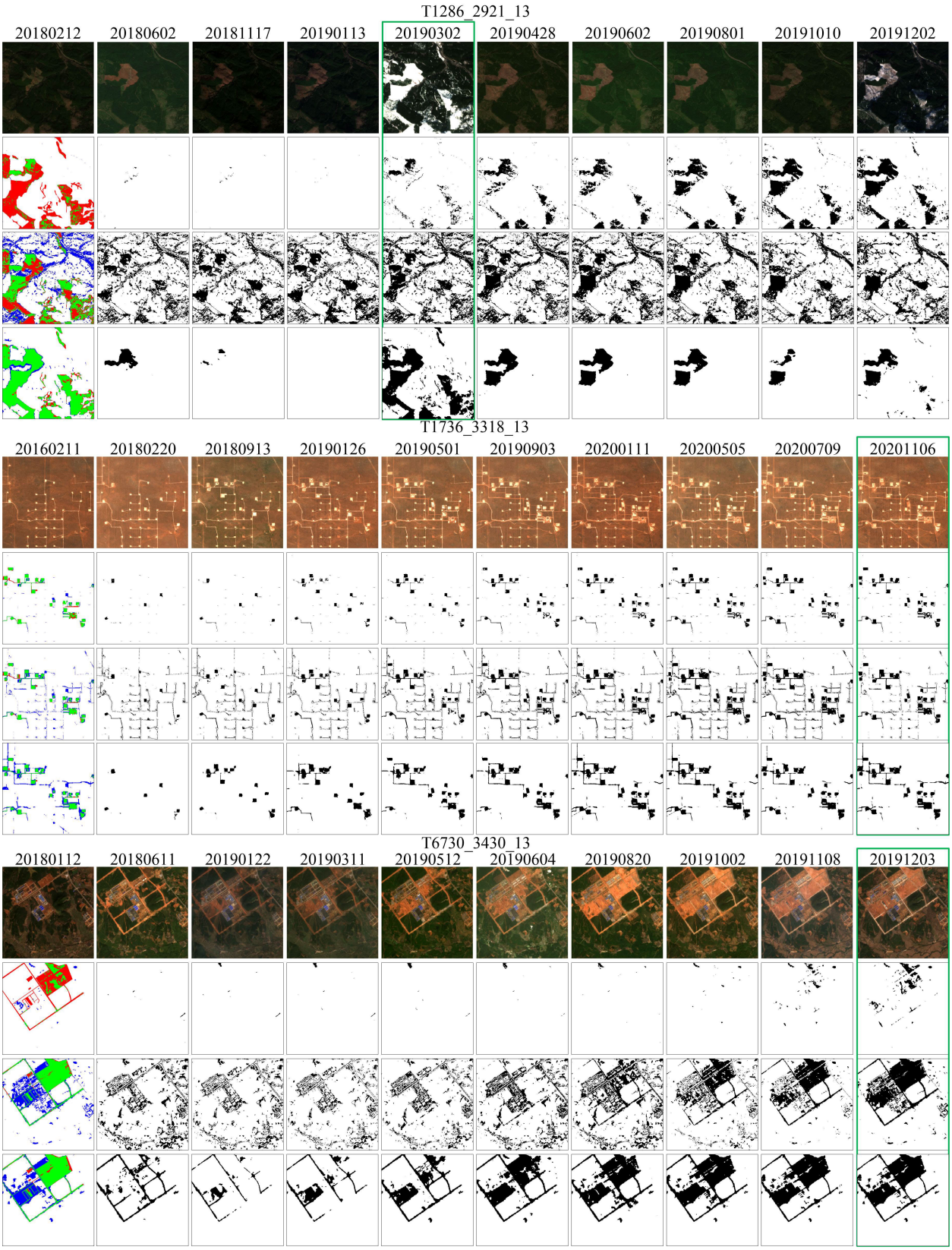


Fig. 3. Examples of change detection results on three scenes for the Sentinel-2 dataset. Row 1: image time-series. Row 2: change maps of one-scene fitting obtained by UTRnet. Row 3: change maps on inference setting obtained by UTRnet. Row 4: change maps on inference setting obtained by the proposed approach. Column 1 of Rows 2–4 in each scene is the most significant change map versus the ground truth (green: TP, white: TN, blue: FP, and red: FN). The green box indicates the most significant changed image pair.

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TABLE II

QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE FITTING TEST SET ON THE LANDSAT-8 DATASET

Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
GMCD	38.06	42.17	83.51	0.400	0.305
CAA	61.09	49.14	89.29	0.545	0.485
PixSSL	53.11	76.01	88.07	0.625	0.557
UTRnet (One-Scene)	60.86	62.57	89.66	0.617	0.557
UTRnet (All-Scene)	17.50	**77.19**	48.50	0.285	0.087
Proposed. (All-Scene)	**76.78**	69.72	**93.15**	**0.731**	**0.692**

TABLE III

QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE INFERENCE TEST SET ON THE SENTINEL-2 DATASET

Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
GMCD	35.78	31.52	82.45	0.335	0.235
CAA	63.85	37.16	88.23	0.470	0.409
PixSSL	55.18	**92.97**	88.02	0.693	0.624
UTRnet (One-Scene)	**85.25**	25.17	88.89	0.388	0.347
UTRnet (Inference)	46.64	64.07	84.67	0.540	0.451
Proposed. (Inference)	70.79	90.76	**93.22**	**0.795**	**0.755**

we present the results of one-scene fitting UTRnet and inference on unseen scenarios based on models trained on all training samples. Similar to the evaluation on the Landsat-8 dataset, we only consider the most significant change map in each scene to assess its quantitative performance (Table III). Among all bi-temporal image change detection approaches, the PixSSL obtained a Kappa coefficient of 0.624, which outperforms the results obtained by GMCD and CAA. In image time-series change detection results, as one can observe, the one-scene fitting UTRnet achieves worse results than those obtained on the Landsat-8 dataset, with an OA of 88.89% and a Cohen's kappa score of 0.35. The possible reason is that the Sentinel-2 dataset contains more seasonal changes such as snow. However, its performance is improved when inferred to unseen scenarios. Nevertheless, it still shows significant improvements compared to the all-scene fitting setting on the Landsat-8 dataset, which is largely due to the increased number and diversity of training samples. On the other hand, the inference results obtained by the proposed approach are significantly better than those obtained by UTRnet in both the one-scene fitting and inference on unseen scenarios settings. Across all five performance metrics, the proposed approach achieves the best performance in most cases, except for precision, achieving an OA of 93.22% and a Cohen's kappa score of 0.755. This indicates that the proposed approach not only outperforms the state-of-the-art method UTRnet on trained samples but also on unseen samples. In the proposed approach the improvement is more pronounced when using a larger and more diverse set of training samples. Similar to the experiments on the Landsat-8 dataset, UTRnet underperforms the results of the state-of-the-art bi-temporal image change detection approach (PixSSL), and the proposed image time-series change detection approach further improved the bi-temporal image change detection results.

In addition to the quantitative analysis, we also provide a visual comparison of the most significant change map and the change map time-series obtained by the proposed approach and the UTRnet method in each scene. Fig. 3 shows a comparison of all methods on the Sentinel-2 test set. The true positive, true negative, false positive, and false negative pixels of the significant change map are colored in green, white, blue, and red, respectively. We first analyze the performance of the most significant change map in each scene. As shown in the figure (first column of change maps in each scene), the proposed approach successfully detects most changed pixels and suppresses seasonal noise whereas the results of UTRnet contain more false alarms and missing detections. For the change map time-series, one can see that the results obtained by one-scene fitting UTRnet contain many missed detections in particular related to the cultivated errors in the image sequence. As for the inference results, UTRnet fails to suppress most seasonal noise and presents more false alarms, while getting big improvements in noise reduction compared with its performance on the Landsat-8 dataset. This issue is well addressed in the proposed approach, where abrupt changes and continuous changes are both well detected.

E. Discussion

In this section, we conduct extensive ablation studies on the proposed approach to analyze the contribution of different components. To better understand the proposed approach, we choose the scene T4780_3377_13 with significant vegetation changes over time for visualization. However, the quantitative evaluation was implemented on the selected 53 scenes of the Sentinel-2 dataset as the ablation test set.

1) Pseudo Labels: Many unsupervised change detection approaches employ a thresholding approach for change detection. However, determining a reasonable threshold is often a challenging task. To demonstrate the effectiveness of the proposed pseudo-label generation approach, we present the pseudo labels obtained by thresholding and feature tracking methods, individually. Then, we train the proposed approach using these two sets of pseudo labels. Finally, we evaluate the performance of the trained models on the ablation test set. Fig. 4 shows the details of the pseudo labels and the results obtained by the trained models. As one can see, the thresholding approach produces more false alarms in the pseudo-change maps with the shorter time interval change pair. In contrast, the feature tracking approach can mitigate the effect of this type of seasonal change while maintaining the most significant changes in the change map time-series. Similarly, the model trained on threshold-based labels produces more missed detections due to this type of noise, while the model trained on feature-tracking refined labels further reduces the false alarms. Table IV presents all five metrics on the ablation test set for the two trained models. Threshold-based labels and feature-tracking refined labels present similar performance on the ablation test set. This is because we only considered the most significant changed map in the evaluation. Among these results, the model trained on feature tracking refined labels provides the best result in almost all metrics, including the highest OA of 94.53% and the highest kappa coefficient of 0.669. Compared to the threshold-based labels, the OA and Kappa on the feature tracking refined labels are further improved by about 2% and 0.05. This indicates that the feature tracking method further improves the pseudo

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Fig. 4. Examples of change detection results on the Sentinel-2 ablation test set. Row 1: image time-series. Row 2: pseudo labels obtained by thresholding approach. Row 3: pseudo labels obtained by feature tracking. Row 4: change maps obtained by the proposed approach only using logistic regression. Row 5: change maps obtained by the proposed approach only using contrastive loss. Row 6: change maps obtained by the proposed approach trained on threshold-based pseudo labels. Row 7: change maps obtained by the proposed approach trained on feature tracking-based pseudo labels. Column 1 of Rows 2–7 is the most significant change map versus the ground truth (green: TP, white: TN, blue: FP, and red: FN). The green box indicates the most significant changed image pair.

labels of RS image time-series and thus benefits the self-training. In addition, the performance of the model trained on threshold-based labels even is worse than the one trained on feature-tracking refined labels only using contrastive loss. This demonstrates the significance of the refined pseudo labels within the proposed approach again.

2) Supervised Contrastive Loss and Contrastive Random Walk Loss: To verify the effectiveness of the contrastive loss and contrastive random walk loss, we set up experiments with training on the proposed pseudo labels. These experiments encompassed training with both supervised contrastive loss and contrastive random walk loss, training solely with supervised contrastive loss, and training solely with logistic regression. The same ablation test set is used here. Fig. 4 and Table IV present the results obtained under three different settings. Results show that the supervised contrastive loss and the contrastive random walk loss achieve significant improvements in noise reduction and maintain the consistency of the time-series change maps. The only use of the contrastive loss achieves the OA of 93.3% and the Kappa of 0.638, which are slightly lower than the values obtained by using both loss functions. In addition, the use of both loss functions increases by about 1% and 0.07 on the OA and the Kappa, respectively, with respect to the only use of the logistic regression. This demonstrates that the joint use of supervised contrastive loss and contrastive random walk loss can further improve the performance of the self-training paradigm.

3) Sensitivity Analysis: Considering that the weighted loss term λ is used in feature learning, we adopt LogME [50] to analyze how supervised contrastive loss and contrastive random walk loss affect feature representation learning. LogME is used to assess pretrained models for transfer learning: a pretrained model with a high LogME value is likely to have good transfer performance. Sensitivity analysis on the weight of each loss term is presented in Table V. We observe that LogME increases when λ increases from 0.01 to 0.1, and our approach achieves better performance. This is because the contrastive random walk pushes the network to learn some similarity among change pairs. However, if λ increases further, the performance drops. This is probably because a large λ forces the model to learn semantic similarity instead of changes.

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TABLE IV

QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE SENTINEL-2 TEST SET IN THE ABLATION STUDY

Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
threshold-based	49.59	77.22	91.02	0.604	0.556
feature-tracking	52.53	67.35	91.71	0.606	0.561
only logistic regression	66.46	60.12	93.77	0.631	0.597
only contrastive learning	59.27	78.23	93.30	0.674	0.638
Pro. on thres.	54.35	**82.49**	92.31	0.655	0.614
Pro. on feature.	**68.18**	71.76	**94.53**	**0.699**	**0.669**

TABLE V

USE LOGME TO ASSESS UNSUPERVISED PRETRAINED MODELS

Pre-trained Unet-ConvLSTM	weight term			
	0.01	0.1	1	10
LogME	0.330	**0.356**	0.300	0.203

V. CONCLUSION

In this work, we have proposed a new framework for detecting changes in RS image time-series without any manually annotated training data. Our framework jointly uses an architecture based on Unet and ConvLSTM and adopts a self-training algorithm. We first extract pseudo labels using the feature-tracking method and then further improve the results by training a model from scratch. Feature-tracking approach detects most changes in the RS image time-series, while alleviating the presence of seasonal noise. The proposed self-training algorithm combines the use of supervised contrastive loss, contrastive random walk loss, and logistic regression following the two-stage setting of supervised contrastive learning. This mitigates the effects of the noise in pseudo labels and keeps the consistency of the change map time-series. Our experiments on two different datasets demonstrate the effectiveness of the proposed approach compared to state-of-the-art methods. It is worth noting that the proposed approach can also generalize well to unseen scenarios. Although our method is demonstrated in the context of multispectral images, it can be applied to other sensors, such as synthetic aperture radar and RGB images.

In future work, we plan to extend our method to detect different types of changes using prior information from multispectral images.

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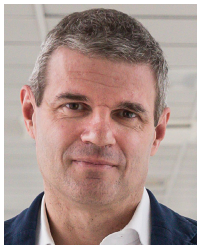
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